

# Limits & Continuity: Guide

**The big idea.** A limit asks where  $f(x)$  is *heading* as  $x$  gets close to  $c$ , not what happens at  $c$  itself. The value  $f(c)$  can be different, or not exist at all. You only need a technique when plugging in fails.

## THE FIVE STEPS

- 1 Plug in first.** Substitute  $x = c$ . If you get a real number, that is your answer (the function is continuous there).

**2 Read what you got.**  $\frac{0}{0}$  means keep working.  $\frac{k}{0}$  with  $k \neq 0$  means a vertical asymptote: check the sign from each side.

**3 Fix 0/0 with algebra.** Factor and cancel, multiply by the **conjugate** when there is a square root, or combine fractions over a common denominator.
- 4 Cancel, then plug in again.** The factor that made the zero is gone, so  $x = c$  now works. Cancelling is legal because a limit never uses  $x = c$  itself.

**5 Check both sides.** The limit exists only if the left-hand and right-hand limits agree. For piecewise functions, always find them separately.

**6 As  $x \rightarrow \pm\infty$ , skip plugging in.** Compare the degrees of the top and bottom (see the degree rule below).

**Most common mistakes:** (1) Writing  $\frac{0}{0} = 0$  or "DNE." It means algebra is needed. (2) Cancelling *terms* instead of *factors*: the  $x$  in  $\frac{x+3}{x}$  cannot be cancelled. (3) Giving a limit at a jump as one of the two values. If left  $\neq$  right, the limit does not exist.

## WHICH TECHNIQUE?

| WHAT YOU GET BY PLUGGING IN                   | WHAT IT MEANS                 | WHAT TO DO                                                                                                      |
|-----------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------|
| A number $L$                                  | Function is continuous at $c$ | <b>Done: the limit is <math>L</math>.</b>                                                                       |
| $\frac{0}{0}$                                 | A hole is hiding there        | <b>Factor and cancel; use the conjugate if there is a radical.</b>                                              |
| $\frac{k}{0}, k \neq 0$                       | Vertical asymptote            | <b>Find each one-sided limit (<math>+\infty</math> or <math>-\infty</math>); if they differ, the limit DNE.</b> |
| $x \rightarrow \pm\infty$ , rational function | End behavior                  | <b>Compare degrees (rule below).</b>                                                                            |
| Left $\neq$ right (piecewise)                 | A jump                        | <b>The two-sided limit DNE.</b>                                                                                 |

## FORMULA BANK

|                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LIMIT LAWS</b><br><br>If $\lim f = L$ and $\lim g = M$ exist:<br><br>Sum / difference $\lim(f \pm g) = L \pm M$<br><br>Product $\lim(fg) = LM$<br><br>Quotient $\lim \frac{f}{g} = \frac{L}{M}, M \neq 0$<br><br>Constant $\lim cf = cL$ | <b>ONE-SIDED LIMITS</b><br><br>$x \rightarrow c^-$ : approach from the <b>left</b> . $x \rightarrow c^+$ : approach from the <b>right</b> .<br><br>$\lim_{x \rightarrow c} f(x) = L$ exactly when $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$ .<br><br>If a one-sided limit is $\pm\infty$ , there is a vertical asymptote at $x = c$ . | <b>DEGREE RULE AT <math>\pm\infty</math></b><br><br>For $\frac{ax^n + \dots}{bx^m + \dots}$ as $x \rightarrow \pm\infty$ :<br><br>$n < m \rightarrow 0$<br><br>$n = m \rightarrow \frac{a}{b}$<br><br>$n > m \rightarrow \pm\infty$<br><br>For $n > m$ , find the sign from $\frac{ax^n}{bx^m} = \frac{a}{b}x^{n-m}$ (an odd power of a negative is negative). |
| <b>CONTINUITY AT <math>x = c</math></b><br><br>$f$ is continuous at $c$ when <b>all three</b> hold:<br>(1) $f(c)$ is defined<br>(2) $\lim_{x \rightarrow c} f(x)$ exists<br>(3) $\lim_{x \rightarrow c} f(x) = f(c)$                        | <b>TYPES OF DISCONTINUITY</b><br><br><b>Removable (hole):</b> the limit exists, but $f(c)$ is missing or different. A factor cancels.<br><br><b>Jump:</b> left and right limits are finite but different.<br><br><b>Infinite:</b> a one-sided limit is $\pm\infty$ . A factor stays in the denominator.                                                         | <b>INTERMEDIATE VALUE THEOREM</b><br><br>If $f$ is <b>continuous</b> on $[a, b]$ and $N$ is between $f(a)$ and $f(b)$ , then $f(c) = N$ for some $c$ in $(a, b)$ .<br><br>To show a root exists: $f(a)$ and $f(b)$ have opposite signs, $N = 0$ .                                                                                                              |

## Seven problems that cover the exam

Same steps every time: plug in, read, fix, check.

## Walkthrough 1: Factor and cancel

0/0 · MOST COMMON

$$\lim_{x \rightarrow 3} \frac{x^2 - 5x + 6}{x - 3}$$

**1 PLUG IN**  $\frac{9 - 15 + 6}{3 - 3} = \frac{0}{0}$ . Indeterminate, so factor.

**2 FACTOR**  $x^2 - 5x + 6 = (x - 2)(x - 3)$

**3 CANCEL**  $\frac{(x - 2)\cancel{(x - 3)}}{\cancel{x - 3}} = x - 2$  for  $x \neq 3$

**4 PLUG IN**  $\lim_{x \rightarrow 3} (x - 2) = \boxed{1}$

## Walkthrough 2: The conjugate

0/0 WITH A RADICAL

$$\lim_{x \rightarrow 4} \frac{x - 4}{\sqrt{x + 5} - 3}$$

**1 PLUG IN**  $\frac{0}{\sqrt{9} - 3} = \frac{0}{0}$ . A radical in the denominator, so use the conjugate.

**2 MULTIPLY** Top and bottom by  $\sqrt{x + 5} + 3$ . The bottom becomes  $(x + 5) - 9 = x - 4$ .

**3 CANCEL**  $\frac{(x - 4)(\sqrt{x + 5} + 3)}{x - 4} = \sqrt{x + 5} + 3$

**4 PLUG IN**  $\sqrt{9} + 3 = 6$   $\boxed{6}$

## Walkthrough 3: Limits at infinity

DEGREE RULE

$$\lim_{x \rightarrow -\infty} \frac{3x^3 - 2x}{5x^3 + x^2 + 7}$$

**1 DEGREES** Top: 3. Bottom: 3. Equal degrees, so the limit is the ratio of the leading coefficients.

**2 CHECK** Divide every term by  $x^3$ . Terms like  $\frac{2}{x^2}$  and  $\frac{7}{x^3}$  go to 0, leaving  $\frac{3}{5}$ .

**3 ANSWER**  $\boxed{\frac{3}{5}}$

## Walkthrough 4: Piecewise continuity

CHECK BOTH SIDES

$$f(x) = \begin{cases} x^2 - 1, & x \leq 2 \\ 2x + 1, & x > 2 \end{cases} \text{ Is } f \text{ continuous at } x = 2?$$

**1 LEFT**  $\lim_{x \rightarrow 2^-} (x^2 - 1) = 3$

**2 RIGHT**  $\lim_{x \rightarrow 2^+} (2x + 1) = 5$

**3 COMPARE**  $3 \neq 5$ , so  $\lim_{x \rightarrow 2} f(x)$  does not exist.

**4 CLASSIFY** Finite left and right limits that differ: **jump discontinuity**. Not continuous at  $x = 2$ . ( $f(2) = 3$ .)

## Walkthrough 5: Holes &amp; asymptotes

RATIONAL FUNCTIONS

$$f(x) = \frac{x(x - 2)}{(x - 2)(x - 5)}. \text{ List every discontinuity and its type.}$$

**1 FIND THEM** The denominator is zero at  $x = 2$  and  $x = 5$ .

**2 CANCEL?**  $(x - 2)$  cancels: **removable** at  $x = 2$ .  $(x - 5)$  stays: **infinite** at  $x = 5$ .

**3 HOLE** Simplify to  $\frac{x}{x - 5}$  and plug in 2:  $\frac{2}{-3}$ . The hole is at  $(2, -\frac{2}{3})$ .

**4 ASYMPTOTE**  $x \rightarrow 5^-: \frac{5}{0^-} = -\infty$ .  $x \rightarrow 5^+: \frac{5}{0^+} = +\infty$ .

## Walkthrough 6: The IVT

SHOW A ROOT EXISTS

Show  $f(x) = x^3 + x - 3$  has a root in  $(1, 2)$ .

**1 CHECK** A polynomial is continuous everywhere, so it is continuous on  $[1, 2]$ .

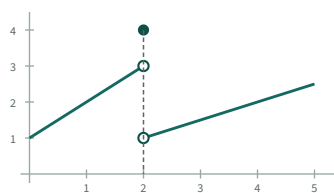
**2 ENDPOINTS**  $f(1) = 1 + 1 - 3 = -1$  and  $f(2) = 8 + 2 - 3 = 7$

**3 CONCLUDE** 0 lies between  $-1$  and  $7$ , so by the IVT there is a  $c$  in  $(1, 2)$  with  $f(c) = 0$ .

**SAY IT** Always name **continuity**, the **two values**, and that the target is **between** them.

## Walkthrough 7: Reading limits from a graph

TRACE WITH YOUR FINGER

The graph of  $f$ , with a jump at  $x = 2$ 

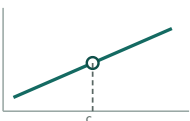
**1 LEFT** Trace toward  $x = 2$  from the left. The height heads to 3 (the open dot), so  $\lim_{x \rightarrow 2^-} f(x) = 3$ .

**2 RIGHT** Trace from the right. The height heads to 1, so  $\lim_{x \rightarrow 2^+} f(x) = 1$ .

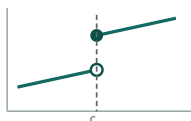
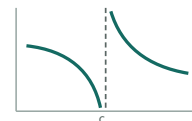
**3 TWO-SIDED**  $3 \neq 1$ , so  $\lim_{x \rightarrow 2} f(x)$  does not exist.

**4 VALUE** The solid dot is the function value:  $f(2) = 4$ . The value can differ from **both** one-sided limits.

## THE THREE DISCONTINUITIES, PICTURED



Removable (hole): the limit exists

Jump: left  $\neq$  right, limit DNE

Infinite: vertical asymptote